

Case Study: Kaymakçı, Turkey

Project Origins: Regional Survey

The Kaymakçı (pronounced “kay-MAK-cha”) Archaeological Project began during an earlier archaeological project, the Central Lydia Archaeological Survey (CLAS). CLAS was a **survey** project, in which archaeologists worked systematically in a bounded geographic area to attempt to locate archaeological material, including **artifacts**, **features**, and **sites**. This survey used both traditional methods of field-walking and innovative technological survey methods involving drone imagery. Kaymakçı is one site identified by CLAS, the largest encountered in the region during the survey and evidently a small city (about 8.5 hectares, roughly the size of Alcatraz Island or 2.5 times the size of Fenway Park), which dates to the Middle and Late Bronze Age (about 2000–1200 BCE).

Intensive pedestrian survey is a method that relies on having individuals precisely spaced at regular intervals walk across a landscape, looking at the ground for evidence of archaeological materials or features. When those materials—such as pottery fragments or stone tools—are found, they are marked, their location and features noted, and they may be collected from the surface. In modern surveying, their precise geographic position is often marked using advanced GPS with spatial precision of one centimeter, which allows these data to be easily digitized.



Intensive pedestrian survey, carrying flags to mark find spots of artifacts.

Source: [Toronto and Region Conservation Authority](https://trca.ca/conservation/archaeology/).

(<https://trca.ca/conservation/archaeology/>)





Lithic artifacts (flaked stone tools) flagged during pedestrian survey in the Western Desert of Egypt.

Source: Dr. Marston

Survey is the systematic search for archaeological remains and sites.

Pedestrian survey uses individuals who are precisely spaced at regular intervals, walking across a landscape and looking at the ground for evidence of archaeological materials or features. In **intensive survey**, these people are closely spaced.

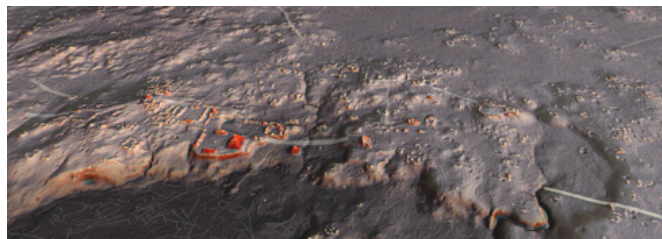
Artifacts are among the most common finds encountered during archaeological survey. As both pottery and stone tools are durable but often break under repeated use, these are the most common types of artifacts to encounter during pedestrian survey.

Features are nonmovable archaeological contexts that were created by people. This is a diverse category that includes, for example, the burned earth remains of a fireplace or clay oven, the foundation stones of a wall, pits or holes dug to store objects or as part of construction, the remains of irrigation canals, and carvings on a bedrock outcrop. Archaeologists often encounter these during excavation, but also

during survey, such as when they find remains of a wall or a pile of stones left at the edge of a field by ancient farmers.

Sites are dense collections of artifacts or features; these can range from an area of the ground with a density of pottery fragments that suggests there may once have been a house there, to the still-standing walls of an ancient city. There are no strict rules that distinguish the minimum size of a site, although individual research teams may create such boundaries when interpreting survey data.

Drone-based imagery allows the detection of a variety of subtle features on the ground that may or may not be visible from the surface. Drone-mounted **lidar** is a new technology that allows precise mapping of ground surfaces by using wavelengths of light that penetrate vegetation. **Photogrammetry** involves taking large numbers of ordinary photographs of three-dimensional surfaces from a variety of angles and stitching them together to create a 2D or 3D virtual rendering of the entire area. Photogrammetry can be of great use in excavation and creating 3D images of artifacts, as well as in survey. **Geophysics** is a powerful set of techniques that use radar, magnetic fields, or electrical currents to detect differences in the composition of the ground across an archaeological site. Geophysics give a map of certain features under the earth to allow archaeologists to see buried walls or burned areas, such as hearths or burned buildings.

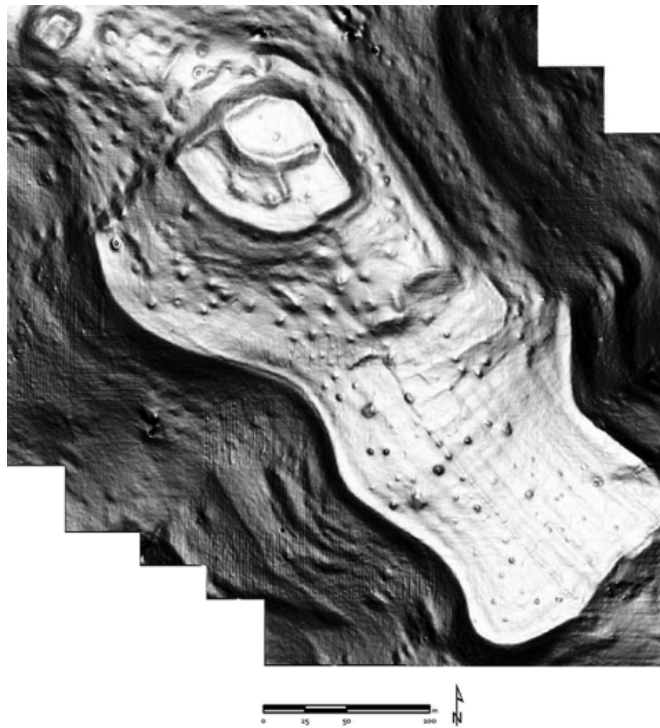


A city imaged with lidar: the Maya site of Naachtun, Guatemala. From overhead, the site is invisible under dense forest.

Source: [Science](https://www.science.org/doi/10.1126/science.aau0137)

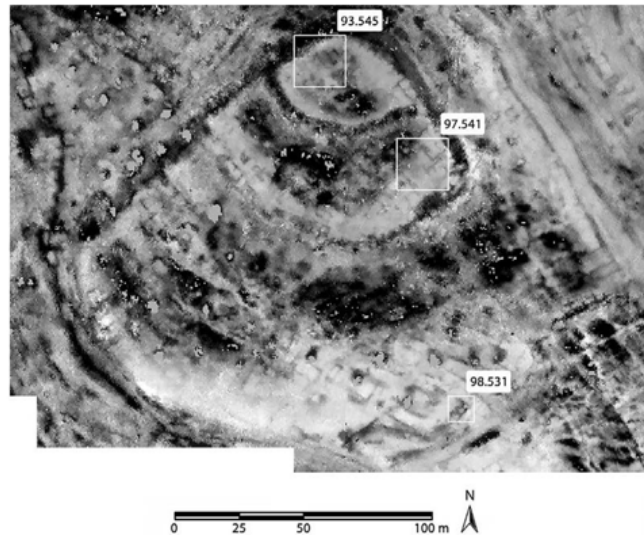
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Subsurface map of Kaymakçı produced using electrical resistivity, a geophysical technique.

Source: [American Journal of Archaeology](https://ajaonline.org/field-report/3764/)
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Lidar is a technology that uses laser beams to scan an area and map point-by-point distances, creating a three-dimensional

rendering of an object, structure, area, or landscape.

Photogrammetry is a computational approach that involves taking large numbers of photographs of three-dimensional surfaces from a variety of angles and stitching them together to create a 2-D or 3-D virtual rendering of the entire area.

Geophysics is a set of techniques that use radar, magnetic fields, or electrical currents to detect differences in the composition of the ground across an archaeological site to map subsurface features.